

Solution Architecture

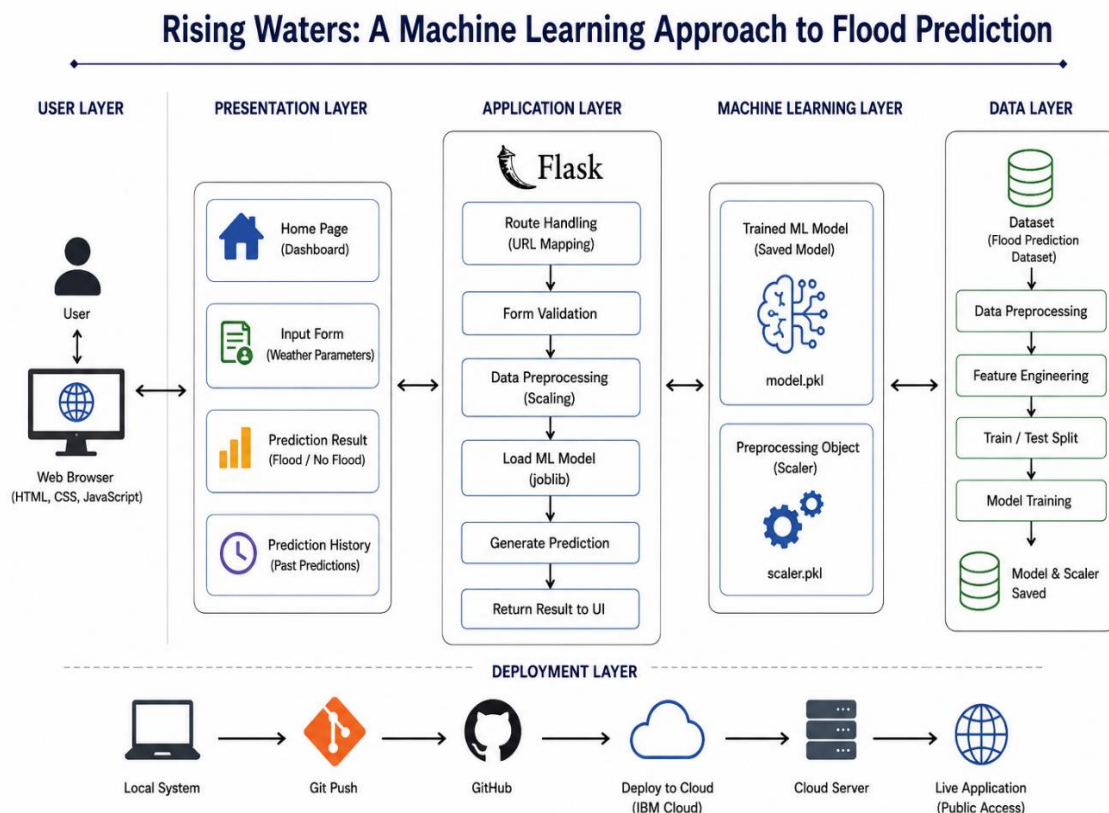
Date	27 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Solution Architecture:

The Solution Architecture for the Rising Waters: A Machine Learning Approach to Flood Prediction project provides an intelligent framework for predicting flood risk using Machine Learning. Its goals are to:

1. Develop an accurate flood prediction system using historical weather data.
2. Define the workflow between the dataset, Machine Learning model, and Flask web application.
3. Process user inputs and generate real-time flood risk predictions.
4. Support scalable deployment on IBM Cloud for reliable and accessible services.

Example - Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter weather parameters, submit prediction requests, view flood prediction results, and access prediction history.	HTML, CSS, JavaScript
API Gateway	Receives requests from the presentation layer, validates user input, preprocesses the data, loads the trained ML model, generates flood predictions, and returns the results to the user interface.	Flask, Python, Joblib
Microservices	Hosts the trained machine learning model and preprocessing (scaler). It processes the input data, performs flood prediction, and sends the prediction result to the API Gateway.	Scikit-learn, Python, model.pkl, scaler.pkl
Database	Stores the flood prediction dataset, supports data preprocessing, feature engineering, train-test splitting, model training, and saves the trained model and scaler files.	Pandas, NumPy, Scikit-learn, Flood Prediction Dataset